# Requirements Brief
Greenie

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## 1 Who we are

We are a start-up company developing a web app that encourages sustainable habits through a gamified eco-action tracker. Students create an account, log their eco-friendly activities and compare their weekly points on a leaderboard, either among all students or among the people they follow.

## 2 Project scope

| | |
|---|---|
| **Project start** | ASAP |
| **Deadline** | Fri 6 Nov 2026 (first milestone: Sat 31 Oct 2026) |
| **Budget** | 100 USD in total, estimated effort 8 to 12 hours |
| **Users** | Students in Zurich, registered with email and password and a profile (nickname, name, date of birth, university) |
| **Platform** | Responsive web app, desktop and mobile layout |
| **Mock-up** | Figma mock-up (Section 11) |
| **Success** | A student can log an action in at most 3 taps; points, streak and the weekly leaderboards (everyone and followed students) update correctly (see Section 8); the code runs from the README and is in a repository we own; the site is deployed on a free hosting tier |

## 3 Deliverables

**Milestone 1**

- **Registration & Login:** Account creation with email and password plus profile fields (nickname, first and last name, date of birth, university) and a consent checkbox; log in and log out on any device (US-01, US-13).
- **Home Status:** Total points, current streak and points earned today (US-02).
- **Action Logging & Point Calculation:** Form interface to log an action from a fixed list of 8 actions; the action's fixed points are added, respecting its daily limit (US-03).
- **Daily Streaks:** Automated tracking of consecutive daily user activity (US-04).
- **Action Configuration File:** Admin control over `actions.json` to define action types, point values and daily limits without code changes (US-08).

**Milestone 2**

- **Find Students & Profiles:** Search for another student by nickname and view their profile (nickname, university, total points, points this week, streak) (US-11).

- **Follow System:** Follow and unfollow other students (US-12).
- **Weekly Leaderboards:** Two rankings based on points from the current week (Monday to Sunday, Zurich time), one for everyone (top 20 plus own rank) and one for the user and the people they follow; points update immediately and the ranking resets weekly (US-06).
- **Personal History (Should):** Individual user log displaying the last 20 logged eco-actions (US-05).
- **Info Guide (Should):** Static "Which bin in Zürich?" page driven by `guide.json`. We supply the content (at least 10 entries) before Milestone 2 starts; you build the page and loading logic only (US-07).
- **Delete My Account (Should):** Removal of the user's account, profile and all related data (US-09).
- **Password Reset (Should):** Reset of a forgotten password through an emailed link (US-10).

**Handover and operations (throughout, final at Milestone 2)**

- **GitHub Repository:** Shared code repository maintained from day one, complete with seed data and a comprehensive `README.md` detailing setup, execution, and instructions for editing `actions.json` and `guide.json`.
- **Live Deployment:** Accessible website URL on a free hosting tier (front end, back end, database and login).
- **AI Governance:** Milestone-by-milestone **AI-use declaration** documenting how AI tools were integrated into the development process.

## 4 Out of scope

Barcode scanning, chatbot, address-based guidance, maps, native apps, photo verification, push notifications, payments, email confirmation at sign-up, sign-in with Google or university accounts, two-factor login, private or blocked profiles, direct messaging.

## 5 Backlog and expectations

**Your role.** Build against the acceptance criteria in Section 8, test, deploy, hand over the repository and declare any AI use.

**Backlog.** The prioritised backlog is Section 6 together with the user stories in Section 8. We work in two sprints that match the two milestones.

**Expectations.** A short written update every two days, early notice if a date is at risk, and a working increment at each milestone. Changes are agreed in writing first.

# 6 Scope and priorities

| Priority | Item |
|---|---|
| Must | Register an account: email, password, unique nickname, first and last name, date of birth, university |
| Must | Log in and log out, on any device |
| Must | Log an action from a fixed list of 8 actions with fixed points and a daily limit |
| Must | Total points, points this week and current daily streak |
| Must | Find a student by nickname and view their profile |
| Must | Follow and unfollow students |
| Must | Weekly leaderboards: everyone (top 20 plus own rank) and the people I follow |
| Must | Action list, points and daily limits editable in one config file, without code changes |
| Must | Runs from README; repository owned by us; deployed on a free tier |
| Should | Own log history (last 20 entries) |
| Should | Static "Which bin in Zürich?" guide, content supplied by us |
| Should | Delete my account and all my data |
| Should | Reset a forgotten password by email |

# 7 Content we supply

## 7.1 Actions

Stored in `actions.json` with the fields `key`, `label`, `points`, `daily_limit`. We supply this file. The values below are the initial content.

| Key | Label | Points | Daily limit |
|---|---|---|---|
| `recycle_bottle` | Recycled a bottle (PET or glass) | 10 | 5 |
| `recycle_can` | Recycled a can | 10 | 5 |
| `recycle_paper` | Recycled paper or cardboard | 10 | 3 |
| `reusable_cup` | Used a reusable cup or bottle | 10 | 3 |
| `reusable_bag` | Used a reusable bag | 5 | 3 |
| `plant_meal` | Ate a plant-based meal | 15 | 3 |
| `green_commute` | Took public transport, bike or walked instead of a taxi or car | 15 | 2 |
| `second_hand` | Bought second-hand or repaired instead of buying new | 20 | 1 |

## 7.2 Guide content (Should)

Stored in `guide.json` with the fields `item`, `where`, `note`. We deliver at least 10 entries before the start of the project. The supplier builds the page and the loading logic only and does not write or change the content.

## 8 User stories and acceptance criteria

Every acceptance criterion (AC) is one pass/fail check at milestone review.

### US-01 Register an account (Must)

*As a new student, I want to create an account and a profile, so that I can log actions, appear on the leaderboards and be found by other students.*

**AC1:** The registration form has seven required fields: email, password (at least 8 characters), nickname (3–20 characters: letters, digits, underscore), first name and last name (1–50 characters each), date of birth (a valid date in the past; I must be at least 16 years old), and university (2–60 characters).
**AC2:** The email must be a valid address and unique, case-insensitive. If it is already registered, an error message appears and no account is created.
**AC3:** Nicknames are unique, case-insensitive. If taken, an error message appears next to the field and no account is created.
**AC4:** Every validation error is shown next to its field. No account is created until all fields are valid.
**AC5:** The form shows a short privacy text (supplied by us) and a consent checkbox. The "Create account" button stays disabled until the box is ticked.
**AC6:** The password field is masked. The password is never stored, logged or sent in plain text.
**AC7:** After success I am logged in and land on Home with 0 points and streak 0.
**AC8:** Email, first name, last name and date of birth are never shown to other users or on any leaderboard. Other users see only nickname, university, points and streak.

### US-02 See my status (Must)

*As a student, I want to see my points and streak, so that I stay motivated.*

**AC1:** Home shows total points, current streak, and points earned today.
**AC2:** Values equal what is stored (the sum of my log entries).
**AC3:** Home has a prominent "Log an action" button.

### US-03 Log an action (Must)

*As a student, I want to log that I did something green, so that I earn points.*

**AC1:** The Log screen lists all actions from `actions.json` with label and points.
**AC2:** Tapping an action adds exactly its points, once per tap.
**AC3:** The success screen shows "+N points", the new total and the streak.
**AC4:** Each action has a daily limit (Europe/Zürich calendar day). Beyond the limit the app shows "Daily limit reached for this action" and adds no points.
**AC5:** Logging works with at most 3 taps from Home.

### US-04 Daily streak (Must)

*As a student, I want a streak, so that I am encouraged to log every day.*

**AC1:** A "day" is a calendar day in Europe/Zürich, midnight to midnight.
**AC2:** The current streak is the number of consecutive days, ending today or yesterday, with at least one logged action.
**AC3:** If nothing was logged yesterday, the streak is 0 until I log again, then 1.

**Test cases** (log history and today's date → expected streak):

| History | Today | Expected streak |
|---|---|---|
| Logged Mon, Tue, Wed | Wed | 3 |
| Logged Mon, Tue, Wed; nothing yet today | Thu | 3 (until end of Thursday) |
| Logged Mon, Tue, Wed; nothing on Thu | Fri | 0 |
| As above, then logs on Fri | Fri | 1 |
| Two actions on the same day | any | counts as one day |
| Action at 23:59 and another at 00:01 | next day | two consecutive days |

### US-05 My history (Should)

*As a student, I want to see my recent logs, so that I can check what I did.*

**AC1:** Lists my last 20 entries: action label, points, local date and time, newest first.
**AC2:** Empty state: "Nothing logged yet."

### US-06 Weekly leaderboards (Must)

*As a student, I want to see how I rank among everyone and among the people I follow, so that I can compare myself with others.*

**AC1:** The Leaderboard screen has two tabs: "Everyone" and "Following".
**AC2:** Both tabs rank by points earned in the current week (Monday 00:00 to Sunday 23:59, Europe/Zürich). A new week starts at 0 points.
**AC3:** "Everyone" shows the top 20: rank, nickname, points. "Following" shows me and every student I follow, in the same columns.
**AC4:** My own row is highlighted. If I am outside the top 20 on "Everyone", my row and rank are shown below the list.
**AC5:** Users with equal points are ordered alphabetically by nickname and get consecutive ranks (no shared ranks).
**AC6:** The leaderboards reflect a new log within 5 seconds of a page refresh.
**AC7:** If I follow nobody, "Following" shows "You are not following anyone yet" and a "Find students" button.
**AC8:** Tapping a row opens that student's profile (US-11).

### US-07 Zürich guide (Should)

*As a student, I want to look up where an item goes, so that I recycle correctly.*

**AC1:** The Guide page lists all entries from `guide.json`.
**AC2:** A text filter narrows the list as I type.
**AC3:** Changing `guide.json` changes the page without code changes.

### US-08 Change the action list without code (Must)

*As the product owner, I want to edit points and actions in one file, so that I do not need change requests.*

**AC1:** Editing `actions.json` (label, points, limit) changes the app after a restart or redeploy, with no code changes.

**AC2:** The README explains how.

### US-09 Delete my account (Should)

*As a student, I want to delete my account, so that my personal data is removed when I leave.*

**AC1:** A "Delete my account" button removes my login, profile data (email, name, date of birth, university, nickname), logs and follow relations after a confirmation.
**AC2:** I disappear from all leaderboards, search results and other students' following lists, and I can no longer log in.

### US-10 Reset my password (Should)

*As a student, I want to reset a forgotten password, so that I do not lose my account.*

**AC1:** "Forgot password?" on the login screen asks for my email and always answers "If this email has an account, we sent a link", whether or not the account exists.
**AC2:** The emailed link lets me set a new password of at least 8 characters. The link expires (the platform's default is acceptable).
**AC3:** After a reset, the new password works and the old one no longer does.
**AC4:** The supplier states the free-tier email sending limits in the bid.

### US-11 Find students and view profiles (Must)

*As a student, I want to find other students and see their profile, so that I can decide whom to follow.*

**AC1:** A search field finds students by nickname (partial match, case-insensitive) and shows up to 20 results with nickname and university.
**AC2:** If nothing matches, the app shows "No student found".
**AC3:** Tapping a result opens the profile: nickname, university, total points, points this week, current streak, and a Follow or Unfollow button.
**AC4:** A profile never shows email, first name, last name or date of birth.

### US-12 Follow and unfollow (Must)

*As a student, I want to follow other students, so that I have a leaderboard with the people I care about.*

**AC1:** The button on a profile toggles between "Follow" and "Unfollow". The state is still correct after a page refresh.
**AC2:** Following is one-way: the other student does not need to approve and is not notified.
**AC3:** I cannot follow myself, and I cannot follow the same student twice.
**AC4:** A new follow appears on my "Following" leaderboard (US-06) within 5 seconds of a page refresh.

### US-13 Log in and log out (Must)

*As a registered student, I want to log in with my email and password on any device, so that I keep my points and follows.*

**AC1:** The login screen has an email field, a masked password field, a "Log in" button, and links to registration and "Forgot password?".
**AC2:** Correct credentials open Home. Wrong credentials show one message, "Email or password is incorrect", without saying which part was wrong.

**AC3:** Login works on any device or browser, and shows the same data.
**AC4:** I stay logged in for at least 7 days or until I tap "Log out".
**AC5:** After "Log out", protected pages are no longer reachable: opening one redirects to the login screen.
**AC6:** A visitor who is not logged in sees only the login and registration screens.
**AC7:** Repeated failed logins are rate-limited (the platform's default is acceptable).

## 9 Non-functional requirements

| Area | Requirement |
|---|---|
| Platform | Responsive web app, mobile-first (works from 360 px width), latest Chrome, Safari and Firefox. No native app. |
| Language | UI in English. |
| Time | All "days" use Europe/Zürich. Timestamps are stored in UTC. |
| Performance | Pages load in under 3 s on a normal mobile connection. |
| Data and privacy | Store only: email, password (as a salted hash, never in plain text), nickname, first name, last name, date of birth, university, follow relations, log entries. No third-party trackers. Email, first name, last name and date of birth are never shown to other users. The supplier works with seed data and test accounts only and keeps no copy of real user data after handover. Data is stored in an EU region. The free plan has no backups, so the seed data stays in the repository. |
| Hosting | Free tier only. Running cost 0 USD. The front end is served as static files from our repository (GitHub Pages is acceptable). Back end, database and login run on one free-tier managed service in an account owned by us, without a payment method; the supplier is added as a developer. The supplier names the service in the bid and states what happens after inactivity (pausing, expiry). The README explains how to keep it running and how to restore it. The deployed instance stays reachable until at least 15 Nov 2026. Preferred service: Supabase (free plan), created by us; the supplier may propose an alternative that meets all rules above and must justify it in the bid. |
| Technology | Front end: supplier's choice of mainstream technology; open source preferred. Back end: see Hosting. No paid services or licences. Stack named in the README. |
| Code structure | Front-end and back-end code in separate files or folders (no markup mixed into server logic). |
| Code quality | Block-level comments explaining each module and function. Clear variable names. |
| Security basics | Validate input, parameterised queries, no secrets in the repository. Use the platform's built-in authentication or a standard salted hash (bcrypt or argon2). HTTPS only. |
| Accessibility | Readable contrast, labelled buttons and fields. |
| Generative AI | AI coding assistants allowed only if declared in writing per milestone (tools and purpose). No AI-written content in `guide.json` or `actions.json`. |

## 10 Data model

| Table or file | Fields |
|---|---|
| `users` | id, email (unique, case-insensitive), password_hash, nickname (unique, case-insensitive), first_name, last_name, date_of_birth, university, created_at |
| `follows` | follower_id, followed_id, created_at (each pair at most once; follower and followed must differ) |
| `action_logs` | id, user_id, action_key, points, logged_at_utc, local_date_zurich |
| `actions.json` | key, label, points, daily_limit |
| `guide.json` | item, where, note |

Derived values: total points is the sum of `points` per user. Points this week are the sum of `points` with `local_date_zurich` in the current Monday-to-Sunday week. The streak is computed from `local_date_zurich` as in US-04. The "Following" leaderboard ranks the user plus all `followed_id` values of that user. Email and password hash may live in the platform's authentication service; `users` then references that account.

## 11 Screens (Figma mock-up)

A view-only Figma mock-up accompanies this brief: [Figma link]. Frame size is $390 \times 844$ (mobile). Each frame is named `S1 ...`, `S2 ...` and its description lists the story IDs. The bottom tab bar on S2, S3, S5 and S6 has: Home, Log, Leaderboard, Guide.

| Frame | Elements | States shown | Stories |
|---|---|---|---|
| S0 Log in | App name, email field, masked password field, “Log in” button, links “Create account” and “Forgot password?” | Default; error “Email or password is incorrect” | US-13 |
| S1 Register | App name, one-line pitch, fields (email, password, nickname, first name, last name, date of birth, university), privacy text with consent checkbox, “Create account” button | Default; errors “Email already registered”, “Nickname taken”, “Password needs at least 8 characters”, “3–20 characters, letters, digits, _”, “You must be at least 16”; button disabled until consent ticked | US-01 |
| S2 Home | Total points (large), streak with flame icon, “Today: +N”, button “Log an action”, links “My history”, “Find students” and “Log out” | New user (0 and 0); active user | US-02, 04, 05, 11, 13 |
| S3 Log action | Title, list of 8 actions, each with label and “+N” | Default; limit-reached message on one action | US-03 |
| S4 Success | “+N points”, new total, streak, buttons “Back to Home” and “Log another” | Streak increased; limit reached | US-03, 04 |
| S5 Leaderboard | Tabs “Everyone” and “Following”; rows: rank, nickname, points; own row highlighted; own rank row pinned below on “Everyone” | Default; user outside top 20; empty list; “Following” empty with “Find students” button | US-06 |
| S6 Guide | Text filter, list of entries (item, where, note) | Default; no results | US-07 |
| S7 History (Should) | List: label, points, date and time | Default; empty | US-05 |
| S8 Find students | Search field, result list (nickname, university) | Default; results; “No student found” | US-11 |
| S9 Student profile | Nickname, university, total | Not following; following | US-11, 12 |

**Navigation:** S0↔S1; S0→S10; S0, S1→S2 after login or registration; S2→S0 on "Log out"; S2→S3→S4→S2; tab bar between S2, S3, S5, S6; S2→S7; S2→S8; S5 ("Following" empty state)→S8; S8→S9; S5 row→S9.

## 12 Delivery and acceptance

**Commercial terms.** Fixed price, paid on the platform in two equal milestones. The total price is agreed in the contract and stays within the budget stated in the job post.

| Milestone | Content | Target date | Payment |
|---|---|---|---|
| M1 | US-01, 02, 03, 04, 08, 13; repository; deployed skeleton | Sat 31 Oct 2026 | 50% of price |
| M2 | US-06, 11, 12; Should: US-05, 07, 09, 10; final fixes; handover | Fri 6 Nov 2026 | 50% of price |

**Acceptance procedure.** We test every acceptance criterion on the deployed URL within 24 hours of delivery and send a defect list. The supplier fixes defects within 24 hours at no extra charge. Payment is released on acceptance, on the platform only.

**Change requests.** In writing, with price and date agreed before work starts.

**Escalation.** No reply within 24 hours triggers a reminder. No reply within 48 hours means we may end the contract and keep the paid-for work.